WAJID ALI

Senior Software Engineer

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Senior Software Engineer with 5+ years building AI-integrated, production-grade systems across Gulf-region and US-based platforms. Specialized in React.js, Next.js, TypeScript, Express.js, NestJS, Flutter, .NET Core, and Laravel, with hands-on experience shipping multi-tenant SaaS, RAG-based AI assistants, agentic automation workflows, real-time infrastructure, and offline-first mobile apps at scale. Leads a team of 6 developers with full end-to-end ownership: architecture, performance, security-first API design, DevOps, and SEO.

TECHNICAL SKILLS

AI & Automation: Claude Code · OpenAI API · RAG · Text-to-SQL · n8n · Nango

Frontend: React 18 · Next.js 16 · Angular · Vite · Redux · TanStack Query · Tailwind

Mobile & Cross-Platform: Flutter (iOS · Android · Web) · React Native · sqflite · Offline-First Sync · FCM

Backend: Node.js · Express · NestJS · ASP.NET · Laravel

Cloud & Infra: AWS (S3, EC2) · Vercel · Firebase · Azure Functions · GCP

DevOps: Linux · Nginx · PM2 · Docker · GitHub Actions · Kubernetes

Databases: PostgreSQL · MySQL · MongoDB · MSSQL · Redis · Drizzle · Prisma

Real-Time: Socket.IO · WebSockets · SSE · Bull/BullMQ

Auth & Security: JWT · OAuth2 · OpenID Connect · RBAC · Azure AD B2C

Integrations: Meta Graph API · YouTube Data API · Resend · WhatsApp Business · Google Auth · Sheets & Apps Script

Web Delivery & SEO: Canonical-domain strategy · DNS/TTL cutover · robots & sitemap generation · Search Console

Performance: Server-side pagination · Code splitting · Lazy loading · Query caching · Compression

Languages: TypeScript · JavaScript · Dart · C# · PHP · Python · Go (active)

Testing & QA: Vitest · Jest · SuperTest · Cypress · Playwright · Visual snapshots · Zod · tsc --noEmit gates

Patterns: CQRS · Microservices · Multi-Tenancy · Repository · MVC · Event-Driven

Tools: Git · GitHub CLI · Postman · Swagger · Serilog · SheetJS · jsPDF

CMS: WordPress (architecture, audit, migration)

PROFESSIONAL EXPERIENCE

Software Engineer · BDCC, Saudi Arabia — **Sep 2023 – Present**

multi-tenant SaaS HRMS · React 18 · Express.js · MySQL · Socket.IO · OpenAI API (RAG) · hr.wajidali.dev

- **Shipped a complete AI-powered multi-tenant HRMS SaaS:** architected, built, and deployed a system serving 3,000+ employees across subsidiary companies, with per-company database provisioning and RBAC; merged 315 PRs across 41 server modules in ~10 days during the SaaS rebuild.
- **Team leadership with full ownership:** lead a team of 6 developers, set engineering standards, and review code while staying hands-on across schema, API, frontend, and DevOps.
- **Performance & API optimization:** cut the Employees list response from 4.3 MB to ~2 KB per page and salary sheet load from ~11 s to ~0.3 s (server-side pagination, lean endpoints, DB-side search/sort); shrank the initial JS download from 3.4 MB to ~500 KB with 122 lazy-loaded chunks and caching.
- **Server-side payroll engine:** matched the legacy calculation with 0 differing rows across two real payroll months (2,100+ employees) and fixed a double-counting bug.
- **RAG-based AI HR assistant (OpenAI):** retrieves live DB schema and HR context, turns natural-language HR questions into validated SQL, and answers from live employee data; running in production.
- **Security & quality:** closed 20+ critical vulnerabilities; built ~2,300 automated tests incl. 123 visual snapshots; QA sweeps eliminated 107 bad-input crashes and 28 faulty audit-log writes.
- **Real-time task & communication layer:** Microsoft To-Do-style task management plus group/1:1 chat via Socket.IO, with roles, assignment, status tracking, and push notifications.
- **Native mobile companion (iOS + Android, Flutter):** employee app covering profile, hierarchy, payroll, vacation and loan requests, vehicles, cases, insurance, and chat; offline-first via sqflite with background sync, Google Sign-In, and FCM.
- **25+ integrated production modules:** payroll, leave/vacation, clearance, iqama/visa, contracts, vehicle fleet, project BOQ, cashier, bank loans, audit logs, and more, all live.
- **Full DevOps ownership:** led Windows → Linux migration, configured Nginx + PM2 for 99.9% uptime, cron-automated MySQL backups, and CI/CD via GitHub Actions.
- **Bilingual document engine:** PDF generation (jsPDF + Amiri Arabic RTL font) and bulk Excel import/export (SheetJS) with Hijri calendar compliance, loaded on demand to keep pages light.

Software Engineer · Beam.ai, Remote — **May 2024 – Mar 2025**

AI automation platform · Next.js · NestJS · Nango · OAuth2

- Built and maintained Nango integration flows for Google Docs, Drive, Sheets, Slack, and additional services, handling OAuth, token refresh, and data sync pipelines.
- Built and maintained the core AI automation platform frontend (Next.js) and backend (NestJS); ensured type-safe, modular service architecture for high-reliability AI workflow delivery.

Software Engineer · Devsinc, Lahore, Pakistan — **Oct 2021 – Aug 2023**

Enterprise .NET/Angular · ASP.NET MVC · .NET Core · Azure AD B2C · CQRS

- Built enterprise apps in ASP.NET MVC, .NET Core Web API, and VB.NET for AfterSchool HQ and iHorizons.
- Implemented OpenID Connect (OpenIDDict) for cross-app SSO; enforced RBAC, admin impersonation, and CORS-compliant REST APIs.
- Architected CQRS with MediatR, Repository + Unit of Work patterns; global exception handling via Serilog; Azure Functions and Laravel-based solutions across multi-stack projects.

SHIPPED PROJECTS

ReelBridge, Multi-Platform Short-Form Video Scheduler · *TypeScript Monorepo · Express · React 18 · PostgreSQL/Drizzle · BullMQ · S3*

- **Bulk cross-posting SaaS:** a creator uploads one batch of clips, fans each out to any mix of connected Facebook Pages, Instagram Business accounts, and YouTube channels with per-platform captions, and schedules the whole set in a single pass. In active development, pre-launch pending Meta App Review and YouTube API compliance audit.
- **Three publishing adapters behind one interface:** Facebook 3-phase /video_reels binary upload, YouTube resumable videos.insert with quota verification and backoff, and Instagram's container-create → poll → publish flow over signed S3 URLs. A shared PlatformAdapter contract keeps each platform a swappable package, so adding TikTok is a new package rather than a schema rewrite.
- **Solved Instagram's missing native scheduling:** Meta offers no future-publish for Reels, so I built an app-managed scheduling worker that wakes at the intended time and runs the full sequence, and surfaced native_scheduled vs awaiting_app_managed_publish as distinct states in the UI, because the two carry different reliability guarantees and users deserve to see which one they're relying on.
- **Rate limiting keyed per platform, not globally:** Instagram's 100-posts/24h, YouTube's quota bucket, and Facebook's Graph limits each get an independent limiter on a BullMQ + Redis queue; the worker deploys and scales separately from the API.
- **Security and tenancy enforced at the query layer:** AES-256-GCM encryption at rest for every OAuth/page token, httpOnly-cookie JWTs with revocable Postgres-backed refresh tokens, and ownership assertions on every route touching a connection, target, or post, verified by integration tests rather than hidden in the UI.
- **Fail-before-you-schedule ergonomics:** streamed multi-file S3 upload with no memory buffering, ffprobe media inspection that flags aspect-ratio and duration violations up front, and a dry-run batch preview that resolves every caption, schedule slot, and warning before a single row is written. DST-safe cadence math via Luxon, and CI that runs lint → build → typecheck → migrate → test against real Postgres and Redis containers rather than mocks.

Client Marketing Site, Build, Launch & Search Recovery · *Next.js 16 · TypeScript · Vercel · Resend · WhatsApp*

- **Canonical-domain architecture:** enforced a single indexable host at the edge via proxy.ts (Next.js 16's renamed middleware), 308-redirects any non-matching Host while preserving path/query, with *.vercel.app and localhost exempted so PR previews and local dev stay intact. app/robots.ts and app/sitemap.ts derive from NEXT_PUBLIC_SITE_URL, so generated SEO artifacts always track the canonical host.
- **Domain, DNS & secrets:** configured apex A record and www CNAME on Vercel; split environment variables between build-time public config and server-only secrets (Resend API key, RFQ sender/recipient) rather than leaking them to the client bundle.
- **Typed content-data layer:** wired the gallery through a GALLERY array + IMG map + getImage() helper in lib/data.ts instead of hardcoding asset paths in components; audited the client's image drop and caught stock catalog photos and AI-chat screenshots before they reached the live gallery.
- **Lead capture:** Resend transactional email for RFQ submissions plus click-to-chat WhatsApp handoff.
- **Diagnosed a reported outage as a stale index, not a bug:** grepped the codebase for the reported "maintenance" string, found nothing, and confirmed the live site rendered clean, isolating it to a cached pre-migration Google crawl. Resolved via Search Console URL Inspection → Request Indexing and sitemap resubmission.
- **Pre-ship verification discipline:** npx tsc --noEmit and npm run build as commit gates, plus curling next dev's rendered HTML to confirm image srcs actually resolved, browser-free validation before opening a PR.

Truck Dispatching & Logistics App · *Flutter · sqflite · FCM · Offline-First*

- **Field-facing dispatch app for a logistics operator:** order intake and lifecycle tracking, driver assignment, trip records and history, in-app chat, and push notifications.
- **Offline-first by necessity, not preference:** drivers lose connectivity mid-route, so the app reads and writes against a local sqflite store and reconciles with the server on reconnect, orders and messages queue locally and never block on network.

Daily Expense Tracker, Cross-Platform · *Flutter (Android + Web) · Firebase · Google Apps Script*

- Single Flutter codebase shipping to both mobile and web, with Google Sign-In authentication and FCM push notifications for spend reminders.
- Google Sheets + Apps Script as the persistence layer, so non-technical users own and inspect their own data in a familiar spreadsheet instead of a black-box database.

Production WordPress Audit, Versioning & Migration Runbook · *WordPress · MySQL · Git · GitHub CLI*

- **Fingerprinted an undocumented live site from the filesystem alone:** read wp-config.php, wp-includes/version.php, and theme/plugin directories to identify WP version, active theme, and plugin stack with no admin access, and flagged Hostinger mu-plugins and LiteSpeed Cache as hosting-coupled code affecting portability.
- **Secrets caught before first commit:** live DB credentials and WordPress auth salts in wp-config.php; built .gitignore from an explicit threat model (credentials, DB dumps, cache/backup artifacts, logs, .env) rather than a generic template, and untracked already-staged cache directories via git rm --cached.
- **History-preserving reorganization:** restructured the repo with git mv so 18,680 files registered as renames instead of delete+add; private repo bootstrapped straight from the working directory using the GitHub CLI, with private visibility chosen deliberately for client-owned content.
- **Deployment documentation as a product:** split docs by purpose (hosting requirements, step-by-step restore, DNS cutover) and cross-linked them into a coherent path; sequenced a downtime-minimizing cutover, verify on new host, lower TTL, switch DNS, monitor propagation, decommission last.

Express.js Fintech API Boilerplate · *Node.js · Express · Socket.IO · JWT · MySQL · MongoDB · Jest*

- **Production-grade modular boilerplate:** layered architecture, JWT + refresh token auth, HttpOnly cookies, RBAC, rate limiting, Helmet, clustering, Socket.IO rooms, full Jest + SuperTest coverage. github.com/wajidalii/express-production-kit

EDUCATION

BS Computer Science · Information Technology University (ITU), Lahore **2017 – 2021**

Operating Systems · Databases · Algorithms · Data Structures · OOP · Software Design Patterns · Mobile & Game Development